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Metronidazole (systemic): Drug information

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For additional information see "[Metronidazole \(systemic\): Patient drug information](#)" and "[Metronidazole \(systemic\): Pediatric drug information](#)"

For abbreviations, symbols, and age group definitions [show table](#)

Special Alerts

Unapproved Compounding Kits September 2024 ^

The FDA has requested that Azurity Pharmaceuticals stop distributing unapproved drugs marketed as "Unit-of-Use Prescription Compounding Kits," including FIRST Lansoprazole, FIRST Pantoprazole, and FIRST Metronidazole. These kits have not been proven safe and effective and do not qualify for exemptions under the compounding provisions in sections 503A and 503B of the Federal Food, Drug and Cosmetic Act.

Further information may be found at <https://www.fda.gov/drugs/enforcement-activities-fda/fda-notification-regarding-unapproved-drugs-included-kits>.

ALERT: US Boxed Warning

Carcinogenic: ^

Metronidazole has been shown to be carcinogenic in mice and rats. Avoid unnecessary use of metronidazole. Reserve metronidazole for use in the following indications: trichomoniasis, amebiasis, and anaerobic bacterial infections.

Brand Names: US

Flagyl [DSC]; Likmez

Brand Names: Canada

APO-MetroNIDAZOLE; Auro-MetroNIDAZOLE; Flagyl [DSC]; M-Metronidazole; MINT-Metronidazole; PMS-Metronidazole

Pharmacologic Category

Amebicide; Antibiotic, Miscellaneous; Antiprotozoal, Nitroimidazole

Dosing: Adult

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Amebiasis, intestinal or extraintestinal

Amebiasis, intestinal (acute dysentery) or extraintestinal (liver abscess): Oral: 500 to 750 mg every 8 hours for 7 to 10 days followed by an intraluminal agent (eg, paromomycin) ([Ref](#)).

Bacterial vaginosis

Bacterial vaginosis:

Note: Treatment is generally not warranted for patients who are asymptomatic ([Ref](#)).

Initial treatment: **Oral:** 500 mg twice daily for 7 days ([Ref](#)). **Note:** For patients who are pregnant, 250 mg 3 times daily for 7 days is an alternative regimen ([Ref](#)).

Recurrent infection: **Oral:** For multiple disease recurrences, 500 mg twice daily for 7 days in combination with or prior to a multiweek course of boric acid, followed by suppressive topical therapy ([Ref](#)).

Male partner management:

Note: For use in male partners of female patients with confirmed bacterial vaginosis to prevent recurrence ([Ref](#)).

Oral: 400 mg twice daily, in combination with topical clindamycin, for 7 days ([Ref](#)). **Note:** Clinical trials were conducted outside of the United States, where metronidazole is available in 200 and 400 mg tablets.

Balantidiasis

Balantidiasis (alternative agent) (off-label use): Oral: 500 to 750 mg 3 times daily for 5 days ([Ref](#)).

Bite wound infection, prophylaxis or treatment

Bite wound infection, prophylaxis or treatment (animal or human bite)

(alternative agent) (off-label use): Oral, IV: 500 mg every 8 hours as part of an appropriate combination regimen ([Ref](#)). Duration is 3 to 5 days for prophylaxis; for established infection, continue for 1 to 2 days after resolution, typically 5 to 14 days total, although deep or complicated infection may require a longer duration ([Ref](#)).

Clostridioides difficile infection, treatment

Clostridioides difficile infection, treatment (off-label use):

Note: Criteria for disease severity is based on expert opinion and should not replace clinical judgment ([Ref](#)).

Nonsevere (ie, WBC $\leq 15,000$ cells/mm³ and serum creatinine <1.5 mg/dL), initial episode (alternative agent):

Note: For patients who cannot receive oral vancomycin or fidaxomicin ([Ref](#)). Avoid metronidazole in patients who are frail, ≥ 65 years of age, or who develop *C. difficile* infection in association with inflammatory bowel disease ([Ref](#)).

Oral: 500 mg 3 times daily for 10 to 14 days ([Ref](#)).

Fulminant infection (ie, ileus, megacolon, and/or hypotension/shock): IV: 500 mg every 8 hours in combination with oral and/or rectal vancomycin for 10 days; may be extended up to 14 days if patient has improved but has not had symptom resolution ([Ref](#)). For patients with clinical improvement, some experts discontinue metronidazole and use vancomycin monotherapy to complete therapy ([Ref](#)).

Crohn disease

Crohn disease (off-label use):

Management after surgical resection:

Monotherapy: **Oral:** 20 mg/kg/day (in 3 divided doses) **or** 1 to 2 g/day in divided doses for 3 months ([Ref](#)); begin as soon as oral intake is resumed after surgery ([Ref](#)).

Combination therapy: **Oral:** 250 mg 3 times daily ([Ref](#)) **or** 1 to 2 g/day in divided doses for 3 months ([Ref](#)); begin as soon as oral intake is resumed after surgery and administer in combination with a thiopurine (azathioprine or mercaptopurine) or a TNF-alpha inhibitor (eg, adalimumab) ([Ref](#)).

Treatment of simple perianal fistulas (adjunctive agent): **Oral:** 500 mg twice daily for 4 weeks initially; if clinical response (ie, cessation of drainage and closure of fistula), continue at 250 mg 3 times daily for an additional 4 weeks ([Ref](#)) **or** 10 to 20 mg/kg/day in divided doses for 4 to 8 weeks with or without ciprofloxacin ([Ref](#)).

Diabetic foot infection, moderate to severe

Diabetic foot infection, moderate to severe: **Oral, IV:** 500 mg every 8 to 12 hours, as part of an appropriate combination regimen ([Ref](#)). Duration (which may include oral step-down therapy) is usually 2 to 4 weeks in the absence of osteomyelitis ([Ref](#)).

Dientamoeba fragilis infection

***Dientamoeba fragilis* infection (alternative agent) (off-label use):**

Note: For patients unable to receive first-line therapy ([Ref](#)).

Oral: 500 to 750 mg 3 times daily for 10 days ([Ref](#)).

Giardiasis

Giardiasis (alternative agent) (off-label use): **Oral:** 250 mg 3 times daily **or** 500 mg 2 times daily for 5 to 7 days ([Ref](#)).

Helicobacter pylori eradication

***Helicobacter pylori* eradication (off-label use):**

Note: Test of cure is recommended at >4 weeks following treatment completion. If possible, proton pump inhibitors (PPIs) should be stopped at least 2 weeks prior to testing to minimize the risk of false-negative results ([Ref](#)).

Bismuth quadruple regimen (preferred): **Oral:** Metronidazole 500 mg 3 or 4 times daily in combination with either bismuth subsalicylate or bismuth subcitrate,

tetracycline, and a PPI; continue regimen for 14 days ([Ref](#)).

Clarithromycin triple regimen (salvage regimen) (alternative regimen):

Note: Clarithromycin-based regimens should be used **only** in patients with documented susceptibility to clarithromycin on antimicrobial susceptibility testing.

Oral: Metronidazole 500 mg three times daily, in combination with clarithromycin plus a potassium-competitive acid blocker or **high dose** PPI; continue regimen for 14 days ([Ref](#)).

Levofloxacin triple regimen (salvage regimen) (alternative regimen):

Note: Levofloxacin-based regimens should be used only **as a last resort** in patients with documented susceptibility to levofloxacin on antimicrobial susceptibility testing ([Ref](#)).

Oral: Metronidazole 500 mg two to three times daily, in combination with levofloxacin plus a PPI; continue regimen for 14 days ([Ref](#)).

Intra-abdominal infection

Intra-abdominal infection:

Note: Empiric oral regimens may be appropriate for patients with mild to moderate infection. Other patients may be switched from IV to oral therapy at the same dose when clinically improved and able to tolerate an oral diet ([Ref](#)). The addition of metronidazole may not be necessary for uncomplicated biliary infection of mild to moderate severity ([Ref](#)). For acute diverticulitis, some experts suggest deferring antibiotics in otherwise healthy patients who are immunocompetent with mild disease ([Ref](#)).

Oral, IV: 500 mg every 8 to 12 hours as part of an appropriate combination regimen ([Ref](#)). Duration of therapy is 4 to 5 days following adequate source control ([Ref](#)). For diverticulitis or uncomplicated appendicitis managed without intervention, duration varies from 4 to 14 days depending on disease extent, response, and immune status ([Ref](#)); for perforated appendicitis managed with laparoscopic appendectomy, 2 to 4 days may be sufficient ([Ref](#)).

Intracranial abscess

Intracranial abscess (brain abscess, intracranial epidural abscess) : IV: 7.5 mg/kg (usually 500 mg) every 6 to 8 hours (maximum dose: 4 g/day) for 6 to 8 weeks in combination with other appropriate antimicrobial therapy ([Ref](#)). **Note:** May switch IV metronidazole to oral metronidazole at the same dose to complete treatment course ([Ref](#)).

Odontogenic infection

Odontogenic infection (off-label use):

Acute simple gingivitis, plaque-associated: Note: Reserve systemic therapy for patients with rapidly progressive disease, severe pain, or immunocompromising conditions ([Ref](#)).

Oral: 500 mg every 8 hours in combination with penicillin V for 5 to 7 days ([Ref](#)).

Periodontitis, severe, plaque-associated: Oral: 500 mg every 8 hours in combination with amoxicillin for 14 days or until clinical resolution; use in addition to periodontal debridement ([Ref](#)).

Pyogenic odontogenic soft tissue infection (alternative agent): IV, Oral: 500 mg every 8 hours as part of an appropriate combination regimen until resolution, typically for 7 to 14 days. Use in addition to appropriate surgical management (eg, drainage and/or extraction) ([Ref](#)).

Pneumonia, aspiration

Pneumonia, aspiration (alternative agent):

Note: For patients with community-acquired aspiration pneumonia who require hospitalization but are not severely ill or for patients who cannot receive first-line therapy ([Ref](#)).

Oral, IV: 500 mg 3 times daily in combination with an appropriate beta-lactam (eg, oral amoxicillin, IV penicillin, or an IV third-generation cephalosporin) for 7 days ([Ref](#)).

Pouchitis, acute

Pouchitis (post ileal pouch-anal anastomosis), acute (off-label use):

Initial therapy (alternative agent): **Oral:** 500 mg every 12 hours for 14 days ([Ref](#)).

Refractory disease: **Oral:** 500 mg every 12 hours in combination with ciprofloxacin for 28 days ([Ref](#)).

Sexually transmitted infections

Sexually transmitted infections:

Empiric treatment following sexual assault in females (off-label use): **Oral:** 500 mg twice daily for 7 days, as part of an appropriate combination regimen ([Ref](#)).

Pelvic inflammatory disease: **Oral, IV:** 500 mg every 12 hours for 14 days as part of an appropriate combination regimen ([Ref](#)).

Trichomoniasis (index case and sex partner):

Initial treatment:

Females: **Oral:** 500 mg twice daily for 7 days. **Note:** Single 2 g dose is less effective ([Ref](#)), but some experts suggest it as an alternative for those unlikely to complete a multiday regimen ([Ref](#)).

Males: **Oral:** 2 g as a single dose ([Ref](#)). **Note:** Some experts suggest the multiday dosing used for females as an alternative regimen ([Ref](#)).

Refractory infection: **Note:** Patients with suspected reinfection because of reexposure to an untreated partner may receive a regimen used for initial treatment. For others, clinicians may request a kit from the CDC to perform drug-resistance testing ([Ref](#)).

Oral: 2 g once daily for 7 days. Alternatively, 500 mg twice daily for 7 days may be sufficient for patients with refractory infection following a single-dose regimen ([Ref](#)).

Skin and soft tissue infection

Skin and soft tissue infection:

Necrotizing infection (as a component of an appropriate combination regimen) (alternative agent): **IV:** 500 mg every 6 hours. Continue until further debridement is not necessary, patient has clinically improved, and patient is afebrile for 48 to 72 hours ([Ref](#)).

Surgical site infection, incisional (eg, intestinal or GU tract; axilla or perineum), warranting anaerobic coverage: **IV:** 500 mg every 8 hours in combination with other appropriate agents. Duration depends on severity, need for debridement, and clinical response ([Ref](#)).

Surgical prophylaxis

Surgical prophylaxis:

IV: 500 mg within 60 minutes prior to surgical incision in combination with other antibiotics. Considered a recommended agent for select procedures involving the GI tract, urologic tract, or head and neck ([Ref](#)).

Oral:

Colorectal surgical prophylaxis (off-label use): **Oral:** 500 mg or 1 g for 3 doses over 7 to 10 hours, in combination with additional oral antibiotics, starting after mechanical bowel preparation the afternoon/evening before a morning surgery and followed by an appropriate IV antibiotic prophylaxis regimen ([Ref](#)). Regimens may vary; refer to institution-specific protocols.

Uterine evacuation (induced abortion or pregnancy loss) (alternative agent) (off-label use): **Oral:** 500 mg as a single dose 1 hour prior to uterine aspiration ([Ref](#)). **Note:** The optimal dosing regimen has not been established; various protocols are in use ([Ref](#)).

Tetanus

Tetanus (*Clostridium tetani* infection) (off-label use): **Oral, IV:** 500 mg every 6 to 8 hours for 7 to 10 days in combination with supportive therapy ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Decreased kidney function does not impact serum half-life of metronidazole but is associated with accumulation of the active hydroxyl and almost inactive acetic acid metabolites (Ref). Kidney dysfunction may confer a higher risk of CNS side effects, such as neuropathy, encephalopathy, and seizures (Ref).

Note: Single-dose regimens (eg, 2 g once) do not require dose adjustment (Ref).

Altered kidney function:

IV, Oral:

CrCl $\geq$ 10 mL/minute: No dosage adjustment necessary (Ref); however, monitor closely for adverse effects due to accumulation of metabolites in patients with more severe impairment (CrCl <30 mL/minute), particularly with prolonged courses of therapy (Ref).

CrCl <10 mL/minute: No dosage adjustment necessary (Ref); however, monitor closely for adverse effects due to accumulation of metabolites, particularly with prolonged courses of therapy. A dose of 500 mg every 12 hours may be adequate to achieve therapeutic plasma levels for nonsevere non-*Clostridioides difficile* infections (Ref).

Hemodialysis, intermittent (thrice weekly): Dialyzable (42% to 65% metronidazole and hydroxyl and acetic acid metabolites (Ref));

IV, Oral: Usual dose: 500 mg every 8 to 12 hours (Ref). **Note:** Larger doses may be utilized depending on clinical indication, but with close monitoring for adverse effects, particularly if the treatment course is >1 to 2 weeks in duration (Ref). On dialysis days, administer after dialysis. If administration cannot be separated from hemodialysis, consider supplemental dose following hemodialysis.

Peritoneal dialysis: Minimally dialyzed (10% removal with single long dwell (Ref));

IV, Oral: No dosage adjustment necessary (Ref). Monitor for adverse effects due to metabolite accumulation, particularly if the treatment course is >1 to 2 weeks in duration (Ref).

CRRT:

Note: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Recommendations are based on high-flux dialyzers and effluent flow rates of 20 to 25 mL/kg/hour (or ~1,500 to 3,000 mL/hour) unless otherwise noted. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection). Close monitoring of response and adverse reactions (eg, neurotoxicity) due to metabolite accumulation is important:

IV, Oral: No dosage adjustment necessary ([Ref](#)).

PIRRT (eg, sustained, low-efficiency diafiltration):

Note: Drug clearance is dependent on the effluent flow rate, filter type, and method of renal replacement. Appropriate dosing requires consideration of adequate drug concentrations (eg, site of infection). Close monitoring of response and adverse reactions (eg, neurotoxicity) due to drug accumulation is important:

IV, Oral: No dosage adjustment necessary ([Ref](#)). On PIRRT days, administer after PIRRT session. If administration cannot be separated from PIRRT, consider supplemental dose following completion of PIRRT session.

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Metronidazole clearance is decreased and elimination half-life is prolonged in patients with cirrhosis ([Ref](#)). Prolonged use or total cumulative dose may result in accumulation that has been linked to neurotoxicity and encephalopathy, which may be irreversible ([Ref](#)). Metronidazole-induced encephalopathy can usually be confirmed (~90% of cases) through MRI (eg, T2 weighted/coronal fluid-attenuated inversion recovery sequences demonstrating symmetric lesions of the dendritic nuclei) ([Ref](#)). Close monitoring is warranted for early recognition of metronidazole-induced encephalopathy in patients with cirrhosis.

Liver impairment prior to treatment initiation: Single-dose regimens (eg, 2 g once) do not require dose adjustment ([Ref](#)).

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Note: Single-dose regimens (eg, 2 g once) do not require dose adjustment ([Ref](#)).

Child-Turcotte-Pugh class A and B: No dosage adjustment necessary ([Ref](#)).

Child-Turcotte-Pugh class C: Oral, IV:

Capsules:

Amebiasis: 375 mg 3 times daily ([Ref](#)).

Trichomoniasis: 375 mg once daily ([Ref](#)).

Injection, tablets:

If usual recommended frequency is every 12 hours: No adjustment necessary.

If usual recommended frequency is every 6 to 8 hours: Maintain dose but reduce frequency to every 12 hours ([Ref](#)). For example, if usual recommended dose is 500 mg every 6 hours, then reduce to 500 mg every 12 hours.

Oral suspension: Reduce dose by 50% ([Ref](#)).

Dosing: Adjustment for Toxicity: Adult

Severe cutaneous adverse reactions (eg, acute generalized exanthematous pustulosis, drug reactions with eosinophilia and systemic symptoms, Stevens-Johnson syndrome, toxic epidermal necrolysis): Discontinue metronidazole immediately and institute appropriate therapy.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "[Metronidazole \(systemic\): Pediatric drug information](#)")

General dosing:

Infants, Children, and Adolescents:

Oral: 15 to 50 mg/kg/**day** in divided doses every 8 hours; maximum daily dose: 2,250 mg/**day** ([Ref](#)).

IV: 22.5 to 40 mg/kg/**day** in divided doses every 6 or 8 hours; maximum daily dose: 4,000 mg/**day** ([Ref](#)).

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Amebiasis



Amebiasis: Infants, Children, and Adolescents: Oral, IV: 35 to 50 mg/kg/**day** in divided doses every 8 hours for 7 to 10 days; maximum dose: 750 mg/dose; followed by an intraluminal agent (eg, paromomycin). Reserve IV therapy for severe infection or extraintestinal disease and switch to oral therapy when tolerated ([Ref](#)).

Bacterial vaginosis



Bacterial vaginosis:

Children <45 kg: Oral: 15 to 25 mg/kg/**day** in divided doses every 8 hours for 7 days; maximum daily dose: 2,000 mg/**day** ([Ref](#)).

Children ≥45 kg and Adolescents: Oral: 500 mg every 12 hours for 7 days ([Ref](#)).

Balantidiasis

Balantidiasis: Infants, Children, and Adolescents: Oral: 35 to 50 mg/kg/**day** in divided doses every 8 hours for 5 days; maximum dose: 750 mg/dose ([Ref](#)).

Clostridioides difficile infection

***Clostridioides difficile* infection:**

Infants: Oral, IV: 7.5 mg/kg/dose every 6 hours for 10 days; maximum dose: 500 mg/dose. If using for severe and complicated disease, use as part of an appropriate combination regimen ([Ref](#)).

Children and Adolescents:

Nonsevere infection, initial or first recurrence: Oral: 7.5 mg/kg/dose every 6 to 8 hours for 10 days; maximum dose: 500 mg/dose ([Ref](#)).

Severe/fulminant infection, initial: IV: 10 mg/kg/dose every 8 hours for 10 days; maximum dose: 500 mg/dose; use concomitantly with oral or rectal vancomycin ([Ref](#)).

Crohn disease, perianal disease; induction

Crohn disease, perianal disease; induction (adjunctive agent): Children and Adolescents: Oral: 20 to 30 mg/kg/**day** in divided doses every 8 to 12 hours as part of an appropriate combination regimen; maximum dose: 500 mg/dose. Reported duration variable; may be continued for ~4 to 12 weeks; overall duration of induction therapy dependent on clinical response ([Ref](#)).

Dientamoeba fragilis

***Dientamoeba fragilis*:** Infants, Children, and Adolescents: Oral: 35 to 50 mg/kg/**day** in divided doses every 8 hours for 10 days; maximum dose: 750 mg/dose ([Ref](#)).

Exit-site or tunnel infection, peritoneal dialysis catheter

Exit-site or tunnel infection, peritoneal dialysis catheter: Infants, Children, and Adolescents: Oral: 10 mg/kg/dose every 8 hours; maximum dose: 500 mg/dose (Ref).

Giardiasis

Giardiasis: Infants, Children, and Adolescents: Oral: 5 mg/kg/dose every 8 hours for 5 to 7 days; maximum dose: 250 mg/dose (Ref).

Helicobacter pylori eradication

***Helicobacter pylori* eradication:** Limited data available:

Note: Use as part of an appropriate combination regimen; optimal duration of therapy is 14 days (Ref).

Weight-directed dosing: Children and Adolescents: Oral: 10 to 15 mg/kg/dose twice daily; maximum dose: 750 mg/dose (Ref).

Fixed dosing:

Children and Adolescents:

15 to <25 kg: Oral: 250 mg twice daily (Ref).

25 to <35 kg: Oral: 500 mg in the morning and 250 mg in the evening (Ref). **Note:** Dose is based on use of tablet dosage form; daily dose may also be split evenly if using suspension (ie, 375 mg twice daily).

35 to 50 kg: Oral: 500 mg twice daily (Ref).

>50 kg: Oral: 750 mg twice daily (Ref).

Intra-abdominal infection

Intra-abdominal infection: Note: Use as part of an appropriate combination regimen. Treatment duration should be limited to 4 to 7 days after source control; longer durations are required when source control inadequate; in some circumstances (eg, acute appendicitis without perforation) therapy should be limited to ≤ 24 hours (Ref).

Divided dosing:

Infants <4 months: IV: 15 mg/kg loading dose once, followed 24 hours later by 7.5 mg/kg/dose every 6 hours ([Ref](#)).

Infants ≥4 months, Children, and Adolescents: IV: 30 to 40 mg/kg/**day** in divided doses every 8 hours; maximum dose: 500 mg/dose ([Ref](#)).

Once-daily dosing: Appendicitis: Children and Adolescents: IV: 30 mg/kg/dose every 24 hours; maximum dose: 1,500 mg/dose ([Ref](#)). For patients weighing <80 kg, some studies report a maximum dose of 1,000 mg/dose ([Ref](#)).

Pelvic inflammatory disease

Pelvic inflammatory disease: Adolescents: Oral, IV: 500 mg every 12 hours for 14 days as part of an appropriate combination regimen ([Ref](#)).

Peritonitis in patients receiving peritoneal dialysis

Peritonitis in patients receiving peritoneal dialysis:

Prophylaxis: GI or GU procedures: Infants, Children, and Adolescents: IV: 10 mg/kg once prior to procedure in combination with cefazolin; maximum dose: 1,000 mg/dose ([Ref](#)).

Treatment: Infants, Children, and Adolescents: Oral: 10 mg/kg/dose every 8 hours; maximum dose: 400 mg/dose ([Ref](#)).

Sexually transmitted infections, empiric treatment following sexual assault in females

Sexually transmitted infections, empiric treatment following sexual assault in females: Adolescent females: Oral: 500 mg every 12 hours for 7 days as part of an appropriate combination regimen ([Ref](#)).

Small intestinal bacterial overgrowth

Small intestinal bacterial overgrowth: Very limited data available: **Note:** Optimal dosing regimen and duration of therapy are unknown; some prescribers report alternating courses of different antibiotics ([Ref](#)). Dosing based on small descriptive studies and published recommendations.

Children and Adolescents: Oral: Usual dose: 10 mg/kg/dose every 12 hours for 7 to 14 days. Maximum dose has not been established for this indication; however,

a maximum of 500 mg/dose has been used for other indications. Reported range: 5 to 10 mg/kg/dose every 8 to 12 hours; may also consider guideline-recommended adult dose of 250 mg every 8 hours in appropriately sized patients ([Ref](#)).

Surgical prophylaxis

Surgical prophylaxis: Children and Adolescents: IV: 15 mg/kg as a single dose 30 to 60 minutes prior to procedure; maximum dose: 500 mg/dose ([Ref](#)).

Surgical prophylaxis, colorectal: Children and Adolescents: Oral: 15 mg/kg/dose every 3 to 4 hours for 3 doses, starting after mechanical bowel preparation the afternoon and evening before the procedure, with or without additional oral antibiotics and with an appropriate IV antibiotic prophylaxis regimen; maximum dose: 1,000 mg/dose ([Ref](#)).

Tetanus

Tetanus (*Clostridium tetani* infection): Infants, Children, and Adolescents: IV, Oral: 30 mg/kg/**day** in divided doses every 8 hours for 7 to 10 days; maximum dose: 500 mg/dose ([Ref](#)).

Trichomoniasis, treatment

Trichomoniasis, treatment:

Children <45 kg: Oral: 45 mg/kg/**day** in divided doses every 8 hours for 7 days; maximum daily dose: 2,000 mg/**day** ([Ref](#)).

Children ≥45 kg and Adolescents:

Females: Oral: 500 mg every 12 hours for 7 days ([Ref](#)).

Males: Oral: 2,000 mg as a single dose ([Ref](#)).

Ulcerative colitis, pouchitis

Ulcerative colitis, pouchitis: Children and Adolescents: Oral: 20 to 30 mg/kg/**day** in divided doses every 8 hours for 14 days; maximum dose: 500 mg/dose ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents:

Manufacturer's labeling:

Mild, moderate, or severe impairment: There are no dosage adjustments provided in the manufacturer's labeling; however, decreased renal function does not alter the single-dose pharmacokinetics.

ESRD requiring dialysis: Metronidazole metabolites may accumulate; monitor for adverse events; accumulated metabolites may be rapidly removed by dialysis.

Intermittent hemodialysis (IHD): If administration cannot be separated from hemodialysis, consider supplemental dose following hemodialysis.

Peritoneal dialysis (PD): No dosage adjustment necessary.

Alternate dosing: Others have used the following adjustments ([Ref](#)). **Note:** Renally adjusted dose recommendations are based on doses of 15 to 30 mg/kg/**day** divided every 6 to 8 hours.

GFR ≥ 10 mL/minute/1.73 m²: No adjustment required.

GFR < 10 mL/minute/1.73 m²: 4 mg/kg/dose every 6 hours.

Intermittent hemodialysis (IHD): Extensively removed by hemodialysis: 4 mg/kg/dose every 6 hours.

Peritoneal dialysis (PD): Extensively removed by peritoneal dialysis: 4 mg/kg/dose every 6 hours.

Continuous renal replacement therapy (CRRT): No adjustment required.

Dosing: Liver Impairment: Pediatric

Infants, Children, and Adolescents:

Mild or moderate impairment: No dosage adjustment necessary; use with caution and monitor for adverse events.

Severe impairment: Reduce dose by 50% ([Ref](#)). Based on experience in adult patients, the dosing interval may be prolonged while maintaining the usual individual dose (eg,

administration of the usual dose but scheduling it every 12 hours instead of every 6) (Ref).

Adverse Reactions (Significant): Considerations

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CNS effects

Metronidazole has been associated with a range of CNS effects, which include **peripheral neuropathy** (Ref), **aseptic meningitis** (Ref), **ataxia** (Ref), **neuro cerebellar toxicity** (Ref), **confusion** or disorientation (Ref), **dysarthria** (Ref), **encephalopathy** (Ref), **seizures** (Ref), **optic neuropathy** (Ref), and **vertigo** (Ref). Most reports have occurred in adults; however, there are reports in pediatric patients (Ref). CNS effects are generally reversible within days to weeks of discontinuation of therapy (Ref); however, some cases may not be reversible (Ref). Peripheral neuropathy symptoms may be prolonged after discontinuation.

Mechanism: Unclear; proposed mechanisms include binding to neural RNA inhibiting protein synthesis, modulation of inhibitory neurotransmitters gamma-aminobutyric acid receptors within the cerebellar and vestibular systems, reversible mitochondrial dysfunction, and vasogenic and cytotoxic edema (Ref).

Onset: Varied; median of 28 days was reported in a systematic review of metronidazole-induced encephalopathy (Ref) and a median of 54 days was reported in another systematic review (Ref); however, both reviews reported that some cases took less than a week.

Risk factors:

May include:

- High doses, high cumulative doses, or prolonged or repeated courses (Ref)

Disulfiram-like reaction

Limited cases of **disulfiram-like reactions** with metronidazole have been reported in both adult and pediatric patients when used concurrently with alcohol (Ref). Available clinical data demonstrating an association are conflicting and the CDC sexually transmitted infections guidelines state that refraining from alcohol use during metronidazole therapy is unnecessary (Ref). The proposed mechanism of disulfiram-like reactions is inhibition of aldehyde dehydrogenase, which results in

the accumulation of aldehydes ([Ref](#)). In one case report, onset was rapid, occurring after 2 days ([Ref](#)). No risk factors have been identified.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Gastrointestinal: Nausea (10% to 12%)

Genitourinary: Vaginitis (15%)

Nervous system: Headache (18%)

1% to 10%:

Gastrointestinal: Abdominal pain (4%), diarrhea (4%), metallic taste (9%), xerostomia (2%)

Genitourinary: Dysmenorrhea (3%), genital pruritus (5%), urinary tract infection (2%), urine abnormality (3%)

Infection: Bacterial infection (7%), candidiasis (3%)

Nervous system: Dizziness (4%)

Respiratory: Flu-like symptoms (6%), pharyngitis (3%), rhinitis (4%), sinusitis (3%), upper respiratory tract infection (4%)

Frequency not defined:

Cardiovascular: Chest pain, flattened T-wave on ECG, flushing, palpitations, peripheral edema, syncope, tachycardia

Dermatologic: Erythematous rash, hyperhidrosis, pruritus, urticaria

Endocrine & metabolic: Decreased libido

Gastrointestinal: Abdominal cramps, anorexia, constipation, decreased appetite, dysgeusia, epigastric discomfort, glossitis, hairy tongue, proctitis, stomatitis, vomiting

Genitourinary: Cystitis, dyspareunia, dysuria, urinary incontinence, vaginal dryness, vulvovaginal candidiasis

Hematologic & oncologic: Agranulocytosis, eosinophilia

Hypersensitivity: Facial edema

Local: Inflammation at injection site, injection-site reaction

Nervous system: Asthenia, chills, depression, drowsiness, hypoesthesia, insomnia, irritability, malaise, numbness, psychosis

Neuromuscular & skeletal: Arthralgia, muscle spasm, myalgia

Ophthalmic: Abnormal eye movements (saccadic), nystagmus disorder

Renal: Polyuria

Respiratory: Dyspnea, nasal congestion

Miscellaneous: Fever

Postmarketing:

Cardiovascular: Prolonged QT interval on ECG ([Ref](#))

Dermatologic: Acute generalized exanthematous pustulosis, bullous dermatitis, fixed drug eruption, Stevens-Johnson syndrome ([Ref](#)), toxic epidermal necrolysis ([Ref](#))

Gastrointestinal: Pancreatitis ([Ref](#))

Genitourinary: Urine discoloration ([Ref](#))

Hematologic & oncologic: Leukopenia ([Ref](#)), neutropenia ([Ref](#)), thrombocytopenia ([Ref](#))

Hepatic: Acute hepatic failure (especially in Cockayne syndrome) ([Ref](#)), hepatotoxicity (including severe hepatotoxicity, especially in Cockayne syndrome) ([Ref](#))

Hypersensitivity: Anaphylaxis ([Ref](#)), drug reaction with eosinophilia and systemic symptoms, serum sickness-like reaction (joint pains) ([Ref](#))

Nervous system: Aseptic meningitis ([Ref](#)), ataxia ([Ref](#)), confusion ([Ref](#)), dysarthria ([Ref](#)), encephalopathy ([Ref](#)), mania ([Ref](#)), neurocerebellar toxicity ([Ref](#)), paresthesia ([Ref](#)), peripheral neuropathy ([Ref](#)), seizure ([Ref](#)), vertigo ([Ref](#))

Ophthalmic: Optic neuropathy ([Ref](#))

Otic: Auditory impairment, hearing loss (including sensorineural hearing loss) ([Ref](#)), tinnitus

Miscellaneous: Disulfiram-like reaction (with alcohol) ([Ref](#))

Contraindications

Hypersensitivity to metronidazole, nitroimidazole derivatives, or any component of the formulation; during the first trimester of pregnancy in patients with trichomoniasis (with the exception of Likmez); use of disulfiram within the past 2 weeks; use of alcohol or propylene glycol-containing products during therapy or within 3 days of therapy discontinuation; Cockayne syndrome.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Note: Although the manufacturer's labeling lists use of alcohol-containing products during therapy or within 3 days of therapy discontinuation as a contraindication, the CDC sexually transmitted infection guidelines state refraining from alcohol use while taking metronidazole is not necessary (CDC [Workowski 2021]). Clinical data demonstrating an association between concomitant use with alcohol and a disulfiram-like reaction are conflicting (Mergenhagen 2020).

Canadian labeling: Additional contraindications (not in the US labeling): Active neurological disorders; history of blood dyscrasia; hypothyroidism; hypoadrenalism.

Warnings/Precautions

Concerns related to adverse effects:

- Superinfection: Prolonged use may result in fungal or bacterial superinfection, including *C. difficile*-associated diarrhea (CDAD) and pseudomembranous colitis; CDAD has been observed >2 months postantibiotic treatment.

Disease-related concerns:

- Hepatic impairment: Use with caution in patients with hepatic impairment due to potential accumulation; dosage adjustment recommended in patients with severe hepatic impairment.
- Renal impairment: Use with caution in patients with renal impairment due to potential accumulation; dosage adjustments may be required.
- Seizure disorder: Use with caution in patients with a history of seizure disorder.

Dosage-form specific issues:

- Injection: Use injection with caution in patients with heart failure, edema, or other sodium-retaining states, including corticosteroid treatment due to high sodium content. In patients receiving continuous nasogastric secretion aspiration, sufficient metronidazole may be removed in the aspirate to cause a reduction in serum levels.

Dosage Forms Considerations

Parenteral solution contains 28 mEq of sodium/gram of metronidazole.

First-Metronidazole oral suspension is a compounding kit. Refer to manufacturer's labeling for compounding instructions.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral:

Flagyl: 375 mg [DSC]

Generic: 375 mg

Solution, Intravenous:

Generic: 500 mg (100 mL); 500 mg/100 mL (100 mL)

Solution, Intravenous [preservative free]:

Generic: 500 mg (100 mL); 500 mg/100 mL (100 mL)

Suspension, Oral:

Likmez: 500 mg/5 mL (75 mL, 200 mL) [contains methylparaben (methyl p-hydroxybenzoate), propylparaben (propyl p-hydroxybenzoate); strawberry peppermint flavor]

Tablet, Oral:

Generic: 125 mg, 250 mg, 500 mg

Generic Equivalent Available: US

May be product dependent

Pricing: US

Capsules (metroNIDAZOLE Oral)

375 mg (per each): \$12.29

Solution (metroNIDAZOLE Intravenous)

500 mg/100 mL (per mL): \$0.02 - \$0.04

Suspension (Likmez Oral)

500 mg/5 mL (per mL): \$3.27

Tablets (metroNIDAZOLE Oral)

125 mg (per each): \$16.59

250 mg (per each): \$0.43 - \$0.48

500 mg (per each): \$0.73 - \$0.84

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral:

Flagyl: 500 mg [DSC] [contains fd&c blue #1 (brilliant blue), quinoline yellow (d&c yellow #10)]

Generic: 500 mg

Solution, Intravenous:

Generic: 500 mg/100 mL (100 mL)

Tablet, Oral:

Generic: 250 mg

Extemporaneous Preparations

Note: A metronidazole suspension is commercially available as 500 mg/5 mL (200 mL bottle) as well as 50 mg/mL or 100 mg/mL compounding kit.

50 mg/mL Oral Suspension (ASHP Standard Concentration) ([Ref](#))

A 50 mg/mL oral suspension may be made with tablets and a 1:1 mixture of Ora-Sweet and Ora-Plus. Crush twenty-four 250 mg tablets in a mortar and reduce to a fine powder. Add small portions of the vehicle and mix to a uniform paste; mix while adding the vehicle in incremental portions to **almost** 120 mL; transfer to a calibrated bottle, rinse mortar with vehicle, and add quantity of vehicle sufficient to make 120 mL. Label "shake well". Stable for 60 days at room temperature or refrigerated ([Ref](#)).

Administration: Adult

IV: Infuse intravenously over 30 to 60 minutes. Avoid contact of drug solution with equipment containing aluminum.

Oral: Administer with food to minimize stomach upset. Shake suspension well before use; administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose.

Administration: Pediatric

Oral: May administer with food to minimize GI upset.

Suspension: Shake well before use. Administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose.

Parenteral: IV: Administer undiluted (5 mg/mL) by slow intermittent infusion over 30 to 60 minutes. Avoid contact of drug solution with equipment containing aluminum.

Storage/Stability

Capsule, tablet: Store at 15°C to 25°C (59°F to 77°F). Protect the tablets from light.

Injection: Store at 20°C to 25°C (68°F to 77°F). Protect from light. Avoid excessive heat. Do not refrigerate. Do not remove unit from overwrap until ready for use. Discard unused solution.

Oral suspension: Store at 20°C to 25°C (68°F to 77°F); excursion permitted between 15°C to 30°C (59°F to 86°F). Do not freeze. Discard any unused suspension 60 days after first opening.

Use: Labeled Indications

Amebiasis: Treatment of acute intestinal amebiasis (amebic dysentery) and extraintestinal amebiasis (liver abscess).

Limitations of use: When used for amebic liver abscess, may be used concurrently with percutaneous needle aspiration when clinically indicated.

Anaerobic bacterial infections (caused by *Bacteroides* spp. , including the *B. fragilis* group):

Bone and joint infections: Treatment (adjunctive therapy) of bone and joint infections.

CNS Infections: Treatment of CNS infections, including meningitis and brain abscess.

Endocarditis: Treatment of endocarditis.

Gynecologic infections: Treatment of gynecologic infections including endometritis, endomyometritis, tubo-ovarian abscess, and postsurgical vaginal cuff infection (also caused by *Clostridium* spp., *Peptococcus* spp., *Peptostreptococcus* spp., and *Fusobacterium* spp.).

Intra-abdominal infections: Treatment of intra-abdominal infections, including peritonitis, intra-abdominal abscess, and liver abscess (also caused by *Clostridium* spp., *Eubacterium* spp., *Peptococcus* spp., and *Peptostreptococcus* spp.).

Lower respiratory tract infections: Treatment of lower respiratory tract infections, including pneumonia, empyema, and lung abscess.

Sepsis: Treatment of sepsis including bloodstream infection (also caused by *Clostridium* spp.).

Skin and skin structure infections: Treatment of skin and skin structure infections (also caused by *Clostridium* spp., *Peptococcus* spp., *Peptostreptococcus* spp., and *Fusobacterium* spp.).

Surgical prophylaxis (colorectal surgery): Injection: Preoperative, intraoperative, and postoperative prophylaxis to reduce the incidence of postoperative infection in patients undergoing elective colorectal surgery classified as contaminated or potentially contaminated.

Trichomoniasis: Treatment of infections caused by *Trichomonas vaginalis*, including treatment of asymptomatic sexual partners.

Use: Off-Label: Adult

Balantidiasis; Bite wound infection, prophylaxis or treatment (animal or human bite); Clostridioides difficile infection, treatment; Crohn disease; Dientamoeba fragilis infection; Empiric treatment (of sexually transmitted infections) in females following sexual assault; Giardiasis; Helicobacter pylori eradication; Odontogenic infection; Pouchitis (post ileal pouch-anal anastomosis), acute; Surgical prophylaxis, colorectal surgery (oral metronidazole); Surgical prophylaxis, uterine evacuation (induced abortion or pregnancy loss); Tetanus

Medication Safety Issues

[Collapse All](#)

Sound-alike/look-alike issues:

MetroNIDAZOLE may be confused with mebendazole, meropenem, metFORMIN, methotrexate, metoclopramide, miconazole.

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drugs (contraindicated in pregnancy [first trimester in patients with trichomoniasis, except Likmez]; pediatric liquid medications requiring measurement) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021).

Metabolism/Transport Effects

Substrate of CYP2A6 (Major with inducers), CYP2A6 (Minor with inhibitors); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential;

Inhibits CYP2C9 (Weak)

Drug Interactions

(For additional information: [Launch drug interactions program](#))

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Aescin: Antibiotics may increase serum concentrations of Aescin. Management: Consider alternatives to this combination as aescin product labeling states coadministration with antibiotics is not recommended. *Risk D: Consider Therapy Modification*

Bacillus Calmette Guerin (BCG) (Intravesical): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Intravesical). *Risk X: Avoid*

Bacillus Calmette Guerin (BCG) (Percutaneous): Antibiotics may decrease therapeutic effects of Bacillus Calmette Guerin (BCG) (Percutaneous). *Risk C: Monitor*

Bacillus clausii: Antibiotics may decrease therapeutic effects of Bacillus clausii. Management: Bacillus clausii should be taken in between antibiotic doses during concomitant therapy. *Risk D: Consider Therapy Modification*

Busulfan: MetroNIDAZOLE (Systemic) may increase serum concentrations of Busulfan.

Management: Avoid coadministration metronidazole and busulfan due to increased risks of busulfan toxicity. If coadministration is required, monitor busulfan concentrations closely and adjust the busulfan dose as needed. *Risk D: Consider Therapy Modification*

Cholera Vaccine: Antibiotics may decrease therapeutic effects of Cholera Vaccine.

Management: Avoid cholera vaccine in patients receiving systemic antibiotics, and within 14 days following the use of oral or parenteral antibiotics. *Risk X: Avoid*

Disulfiram: May increase adverse/toxic effects of MetroNIDAZOLE (Systemic). In particular, the risk for CNS toxicities such as psychosis may be increased. *Risk X: Avoid*

DroNABinol: MetroNIDAZOLE (Systemic) may increase adverse/toxic effects of DroNABinol. Specifically, metronidazole may produce severe intolerance to the alcohol contained in the dronabinol oral solution. *Risk X: Avoid*

Fecal Microbiota (Live) (Oral): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Oral). *Risk X: Avoid*

Fecal Microbiota (Live) (Rectal): Antibiotics may decrease therapeutic effects of Fecal Microbiota (Live) (Rectal). *Risk X: Avoid*

Fluorouracil Products: MetroNIDAZOLE (Systemic) may increase serum concentrations of Fluorouracil Products. *Risk C: Monitor*

Fosphenytoin: May decrease serum concentrations of MetroNIDAZOLE (Systemic). MetroNIDAZOLE (Systemic) may increase serum concentrations of Fosphenytoin. *Risk C: Monitor*

Haloperidol: QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of Haloperidol. *Risk C: Monitor*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): Antibiotics may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). *Risk C: Monitor*

Lactobacillus and Estriol: Antibiotics may decrease therapeutic effects of Lactobacillus and Estriol. *Risk C: Monitor*

Lithium: MetroNIDAZOLE (Systemic) may increase adverse/toxic effects of Lithium. MetroNIDAZOLE (Systemic) may increase serum concentrations of Lithium. *Risk C: Monitor*

Mebendazole: May increase adverse/toxic effects of MetroNIDAZOLE (Systemic). Particularly the risk for Stevens-Johnson syndrome or toxic epidermal necrolysis may be increased. *Risk X: Avoid*

Mycophenolate: Antibiotics may decrease active metabolite exposure of Mycophenolate. Specifically, concentrations of mycophenolic acid (MPA) may be reduced. *Risk C: Monitor*

PHENobarbital: MetroNIDAZOLE (Systemic) may increase adverse/toxic effects of PHENobarbital. A disulfiram-like reaction may occur if combined with phenobarbital dosage forms that contain propylene glycol or alcohol. PHENobarbital may decrease serum concentrations of MetroNIDAZOLE (Systemic). *Risk C: Monitor*

Phenytoin: MetroNIDAZOLE (Systemic) may increase adverse/toxic effects of Phenytoin. A disulfiram-like reaction may occur if combined with phenytoin dosage forms that contain propylene glycol. Phenytoin may decrease serum concentrations of MetroNIDAZOLE (Systemic). MetroNIDAZOLE (Systemic) may increase serum concentrations of Phenytoin. *Risk C: Monitor*

Primidone: May decrease serum concentrations of MetroNIDAZOLE (Systemic). *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-prolonging Agents (Indeterminate Risk - Caution) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk).
Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Sodium Picosulfate: Antibiotics may decrease therapeutic effects of Sodium Picosulfate.
Management: Consider using an alternative product for bowel cleansing prior to a colonoscopy in patients who have recently used or are concurrently using an antibiotic. *Risk D: Consider Therapy Modification*

Sulfonylureas: MetroNIDAZOLE (Systemic) may increase serum concentrations of Sulfonylureas. *Risk C: Monitor*

Tipranavir: MetroNIDAZOLE (Systemic) may increase adverse/toxic effects of Tipranavir. A disulfiram-like reaction may occur due to the alcohol contained in tipranavir capsules. *Risk C: Monitor*

TOLBUTamide: CYP2C9 Inhibitors (Weak) may increase serum concentrations of TOLBUTamide. *Risk C: Monitor*

Typhoid Vaccine: Antibiotics may decrease therapeutic effects of Typhoid Vaccine. Only the live attenuated Ty21a strain is affected. **Management:** Avoid use of live attenuated typhoid vaccine (Ty21a) in patients being treated with systemic antibacterial agents. Postpone vaccination until 3 days after cessation of antibiotics and avoid starting antibiotics within 3 days of last vaccine dose. *Risk D: Consider Therapy Modification*

Vecuronium: MetroNIDAZOLE (Systemic) may increase neuromuscular-blocking effects of Vecuronium. *Risk C: Monitor*

Vitamin K Antagonists: MetroNIDAZOLE (Systemic) may increase serum concentrations of Vitamin K Antagonists. Management: Consider alternatives to concomitant therapy with these agents. If concomitant therapy cannot be avoided, consider reducing the dose of the vitamin K antagonist and monitor for increased INR/bleeding. *Risk D: Consider Therapy Modification*

Food Interactions

Peak antibiotic serum concentration lowered and delayed, but total drug absorbed not affected.

Pregnancy Considerations

Metronidazole crosses the placenta.

Cleft lip with or without cleft palate has been reported following first trimester exposure to metronidazole; however, most studies have not shown an increased risk of congenital anomalies or other adverse events to the fetus following maternal use during pregnancy. Because metronidazole was carcinogenic in some animal species, concern has been raised whether metronidazole should be used during pregnancy. Available studies have not shown an increased risk of infant cancer following metronidazole exposure during pregnancy; however, the ability to detect a signal for this may have been limited.

Metronidazole pharmacokinetics are similar between pregnant and nonpregnant patients (Amon 1981; Visser 1984; Wang 2011).

Bacterial vaginosis and vaginal trichomoniasis are associated with adverse pregnancy outcomes (such as premature rupture of membranes, preterm delivery). The CDC recommends treatment of pregnant patients with symptomatic bacterial vaginosis; the twice daily dose of oral metronidazole for the treatment of bacterial vaginosis is the same for nonpregnant and pregnant patients; however, a three times daily regimen has also been used. Treatment of pregnant patients with asymptomatic bacterial vaginitis who are at high risk for preterm delivery has had mixed results. Metronidazole is also recommended for the treatment of vaginal trichomoniasis in pregnant patients. Although use of oral metronidazole for vaginal trichomoniasis during the first trimester is contraindicated by the manufacturer, available guidelines note treatment can be given at any stage of pregnancy (CDC [Workowski 2021]).

Metronidazole may also be used for the treatment of giardiasis in pregnant patients (some sources recommend second and third trimester administration only) (Gardner 2001; HHS [OI adult 2020]) and symptomatic amebiasis during pregnancy (HHS [OI adult 2020]; Li 1996).

Short courses may be used for the treatment of pouchitis or perianal disease in pregnant patients with inflammatory bowel disease; not recommended for planned maintenance therapy (Mahadevan 2019). The use of other agents is preferred when treatment is needed for *Clostridioides difficile* during pregnancy (ACG [Kelly 2021]). Consult current recommendations for appropriate use in pregnant patients.

Breastfeeding Considerations

Metronidazole and its active hydroxyl metabolite are present in breast milk at concentrations similar to maternal plasma concentrations.

Data related to the presence of metronidazole in breast milk are available from multiple studies:

- Metronidazole transfer to breast milk was assessed in 3 women 6, 13, and 14 weeks' postpartum. Metronidazole 2 g was given orally as a single dose. Breastfeeding was withheld for 24 hours after the dose and several milk samples were collected during these 24 hours. The highest average milk concentrations occurred 2 to 4 hours after the dose; the highest reported average milk concentration was 45.8 mcg/mL. Average milk concentrations were ~12.6 mcg/mL 12 to 24 hours after the dose. The half-life of metronidazole in breast milk was ~9 to 10 hours (2 patients). The authors estimated the infant dose received via breast milk to be 21.8 mg in the first 24 hours and an additional 3.5 mg in the second day after the dose. They noted that withholding breastfeeding for 12 to 24 hours would significantly decrease metronidazole exposure to the breastfeeding infant (Erickson 1981). Metabolite concentrations were not measured in this study. The relative infant dose (RID) of metronidazole is 13.7% to 22.9% when calculated using the highest average breast milk concentration reported in the study (45.8 mcg/mL), compared to an oral infant therapeutic dose of 30 to 50 mg/kg/day, providing an estimated daily infant dose via breast milk of 6.87 mg/kg/day.
- A review article used information from 4 published studies to calculate an RID for metronidazole and compare it to other antibiotics. Maternal doses of oral metronidazole used in these studies ranged from 200 to 400 mg 3 times daily (Heisterberg 1983; Passmore 1988; Willis 1978) or a 2 g single oral dose (Erickson 1981). Authors of the review calculated the RID of metronidazole to be 11%, providing a median estimated daily infant dose via breast milk of 1.59 mg/kg/day (van Wattum 2019).
- In general, breastfeeding is considered acceptable when the RID is <10%; when an RID is >25%, breastfeeding should generally be avoided (Anderson 2016; Ito 2000). Additional considerations can include the gestational and postnatal age of the infant, the actual amount of milk being ingested (less in the first couple days of life and

when weaning), properties of the specific maternal medication, medical conditions of the infant, and medications the infant is receiving therapeutically.

- Breast milk was also evaluated following administration of metronidazole 500 mg IV 3 times a day for 2 days to patients following cesarean delivery. Sampling occurred 1 to 2 hours after the dose on the second day of therapy. Metronidazole concentrations were 7.3 to 10.1 mcg/mL (breast milk) and 7.7 to 13.1 mcg/mL (maternal plasma) (Geballa-Koukoula 2018).

Loose stools, oral and perianal *Candida* growth, and oral thrush have been reported in breastfeeding infants exposed to metronidazole (Passmore 1988).

Metronidazole and its active metabolite can be detected in the serum of breastfeeding infants (Gray 1961; Heisterberg 1983; Passmore 1988).

A potential for tumorigenicity was observed following chronic oral doses in animal studies; the clinical relevance of this is unclear. According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Alternately, the mother can express and discard milk for 48 hours after the last dose and feed the infant stored human milk or formula.

Some guidelines note if metronidazole is given, breastfeeding should be withheld for 12 to 24 hours after a single 2 g dose (CDC [Workowski 2021]; WHO 2002); alternatively, the mother may pump and discard breast milk for 24 hours after taking the last metronidazole dose. Use of lower maternal doses may provide lower concentrations of metronidazole in breast milk and use can be considered in patients who are breastfeeding (CDC [Workowski 2021]). Use of other agents is preferred when treating breastfeeding patients for diseases such as *Clostridioides difficile* infection, or pouchitis or perianal disease in lactating patients with inflammatory bowel disease (ACG [Kelly 2021]).

Dietary Considerations

Take with food to minimize stomach upset.

Sodium: Injectable dosage form may contain sodium.

Ethanol: According to the manufacturer's labeling, use of ethanol is contraindicated during therapy and for 3 days after therapy discontinuation; however, the CDC sexually transmitted infection guidelines state refraining from alcohol use while taking metronidazole is not necessary (CDC [Workowski 2021]). Clinical data demonstrating an association between concomitant use with alcohol and a disulfiram-like reaction are conflicting (Mergenhagen 2020).

Monitoring Parameters

Monitor CBC with differential at baseline, during, and after prolonged or repeated courses of therapy. Closely monitor older patients and patients with severe hepatic impairment or ESRD for adverse reactions. Neurologic symptoms; observe patients carefully if neurologic symptoms occur and consider discontinuation of therapy. Signs and symptoms of severe cutaneous adverse reactions.

Mechanism of Action

After diffusing into the organism, interacts with DNA to cause a loss of helical DNA structure and strand breakage resulting in inhibition of protein synthesis and cell death in susceptible organisms

Pharmacokinetics (Adult Data Unless Noted)

Absorption: Oral: Well absorbed.

Distribution: To bile, seminal fluid, bone, liver, and liver abscesses, lung and vaginal secretions; crosses blood-brain barrier; saliva and CSF concentrations similar to those in plasma.

V_d :

Preterm neonates (PMA: 24 to 39 weeks): Median: 0.95 L/kg (range: 0.75 to 1.03 L/kg) (Cohen-Wolkowicz 2013).

Children ≥ 9 years and Adolescents ≤ 14 years: 0.64 ± 0.18 L/kg (Amon 1983).

Protein binding: $<20\%$.

Metabolism: Hepatic (30% to 60%) to several metabolites including an active hydroxyl metabolite which maintains activity $\sim 30\%$ to 65% of the parent compound (Lamp 1999).

Half-life elimination:

Neonates < 7 days (Jager-Roman 1982): Within first week of life, more prolonged with lower GA:

GA 28 to 30 weeks: 75.3 ± 16.9 hours.

GA 32 to 35 weeks: 35.4 ± 1.5 hours.

GA 36 to 40 weeks: 24.8 ± 1.6 hours.

Neonates ≥ 7 days: ~ 22.5 hours (Upadhyaya 1988).

Children and Adolescents: 6 to 10 hours (Lamp 1999).

Adults: ~8 hours.

Time to peak, serum: Oral: Capsules, tablets: 1 to 2 hours; oral suspension: 0.25 to 6 hours.

Excretion: Urine (unchanged drug and metabolites: 60% to 80%; ~20% of total as unchanged drug); feces (6% to 15%).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: CrCl $\leq$ 65 mL/minute: Half-life: 18 to 32 hours (hydroxy metabolite [active]) (Lamp 1999).

Hepatic function impairment: Half-life: 18.31 hours (mean) in one study (Lau 1987).

According to Child-Pugh classification (Muscara 1995):

Child-Pugh class A: ~10.7 hours.

Child-Pugh class B: ~13.5 hours.

Child-Pugh class C: ~21.5 hours.

Anti-infective considerations:

Parameters associated with efficacy: Concentration dependent; associated with total AUC₂₄/minimum inhibitory concentration (MIC) (Craig 1998; Ibrahim 2004).

Organism specific: *B. fragilis*: Goal: AUC/MIC $\geq$ 70 (Sprandel 2006).

Expected drug exposure in normal renal function:

C_{max} (peak): Single dose:

Neonates and Infants <8 weeks PNA: IV: 20 mg/kg: 17.7 $\pm$ 3.1 mg/L (Rubenson 1986).

Infants 3 to 10 months of age: IV: 20 mg/kg: 25.3 $\pm$ 4.5 mg/L (Rubenson 1986).

Children $\geq$ 4 years of age and Adolescents $\leq$ 17 years of age: IV: 30 mg/kg (maximum dose: Weight <80 kg: 1,000 mg; Weight >80 kg: 1,500 mg): ~50 mg/L (Child 2019).

Adults: Oral:

250 mg: 6 mg/L (manufacturer's labeling).

500 mg: ~8 to 12 mg/L (Lamp 1999; manufacturer's labeling).

2 g: 40 mg/L (manufacturer's labeling).

C_{max} (peak): Steady state:

Preterm Neonates (GA $\leq$ 32 weeks; PNA $\leq$ 90 days): IV: 15 mg/kg loading dose followed by 7.5 mg/kg every 12 hours (PNA $<$ 14 days) or 24 hours (PNA $\geq$ 14 days): 24.6 mg/L (Cohen-Wolkowicz 2013).

Adults: IV:

500 mg every 8 hours: 22.2 ± 5 mg/L (Sprandel 2004).

1 g once daily: 24.8 ± 6.8 mg/L (Sprandel 2004).

1.5 g once daily: 37.7 ± 10 mg/L (Sprandel 2004).

15 mg/kg loading dose; 7.5 mg/kg every 6 hours: 25 mg/L (manufacturer's labeling).

AUC₀₋₂₄ (total): Single dose:

Children $\geq$ 4 years of age and Adolescents $\leq$ 17 years of age (Child 2019): IV: 30 mg/kg (maximum dose: Weight $<$ 80 kg: 1,000 mg; Weight $>$ 80 kg: 1,500 mg):

4 to 5 years of age: 612 to 732 mg•hour/L.

6 to 9 years of age: 520 to 588 mg•hour/L.

10 to 12 years of age: 419 to 478 mg•hour/L.

13 to 17 years of age: 313 to 369 mg•hour/L.

AUC₀₋₂₄ (total): Steady state:

Adults: IV:

500 mg every 8 hours: 356 ± 68 mg•hour/L (Sprandel 2004).

1 g once daily: 227 ± 57 mg•hour/L (Sprandel 2004).

1.5 g once daily: 338 ± 105 mg•hour/L (Sprandel 2004).

Postantibiotic effect: Varies based on the organism and antimicrobial exposure (concentration and duration):

C. difficile: 1.2 to 1.7 hours (Odenholt 2007).

T. vaginalis: ~6 hours (Nix 1995).

Patient Counseling Points

(For additional information see ["Metronidazole \(systemic\): Patient drug information"](#))

What is this drug used for?

- It is used to treat infections.
- It is used to prevent infections during bowel surgery.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

All products:

- Constipation, diarrhea, stomach pain, upset stomach, throwing up, or decreased appetite
- Stomach cramps
- Metallic taste
- Headache
- Joint pain
- Lowered interest in sex

Infusion:

- Irritation where this drug is given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Redness or white patches in mouth or throat
- Vaginal itching or discharge
- Fast or abnormal heartbeat
- Pain when passing urine or blood in urine
- Passing urine more often
- Pelvic pain
- Ringing in the ears, hearing loss, or any other changes in hearing

- Nervous system problems like burning, numbness, or tingling feeling that is not normal; change in balance or eyesight; dizziness or passing out; headache; not able to sleep; seizures; trouble speaking; or feeling confused, depressed, irritable, tired, or weak
- Aseptic meningitis like a headache, fever, chills, very upset stomach or throwing up, stiff neck, rash, bright lights bother your eyes, feeling sleepy, or feeling confused
- Infection like fever, chills, or sore throat
- Severe skin reactions; red, swollen, blistered, or peeling skin; red or irritated eyes; sores in the mouth, throat, nose, eyes, genitals, or any areas of skin; fever; chills; body aches; shortness of breath; or swollen glands
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations ([show table](#))

(AE) United Arab Emirates: Elyzol | Flagyl | Flazole | Metrolag | Neogyl | Nidazol | Riazole | Supplin; (AR) Argentina: Armax | Bexon | Dinestál | Fada metronidazol | Flagyl | Format | Ginkan | Metronidazol | Metronidazol apolo | Metronidazol biocrom | Metronidazol biol | Metronidazol

fabra | Metronidazol lemax | Metronidazol richet | Metronidazol rivero | Metronidazol ugal | Metronidazol upl | Metronidazol vannier | Ovufem | Solucion de metronidazol; (AT) Austria: Anaerobex | Metronidazol | Metronidazol arcana | Metronidazol b braun | Metronidazol genericon pharma | Metronidazol kabi | Metronidazol Vioser | Metronidazol waldheim | Trichex; (AU) Australia: Flagyl | Metronidazole juno | Metronidazole kabi | Metronidazole medsurge | Metronidazole noridem | Metronidazole sandoz; (BD) Bangladesh: Ametri | Amodis | Amotrex | Antipro | Anzole | Biozyl | Brimazol | Danidazol | Dirozyl | Filmet | Flagyl | Flamyd | G-metronidazole | Gyl | Kilpro | Librazol | M Zed | Menilet | Menol | Metason | Metonid | Metra | Metrion | Metro | Metrobac | Metrogyl | Metromax | Metryl | Neomet | Nidazyl | Onida | Prozol | Sb mezol | Servizol | Silmet; (BE) Belgium: Anaeromet | Flagyl; (BF) Burkina Faso: Biozole | Metronidazole b braun; (BG) Bulgaria: Efloran | Flagyl | Metris | Metronidazole abr; (BR) Brazil: Canderm | Dazolston | Endonidazol | Flagyl | Fresenizol | Helmizol | Hidazol | Meflagin | Metrizol | Metronack | Metronidazol | Metroniflex | Nidazofarma | Odonid | Polibiotic | Tenaflor; (CH) Switzerland: Arilin | Dumozol | Flagyl | Metronidazol B. Braun | Metronidazol mepha | Metronidazol zentiva | Metronidazole Bioren | Metronidazole sintetica | Metronidazole vifor medical; (CI) Côte d'Ivoire: Biozole | Metrocare | Metrogyl | Midazole; (CL) Chile: Aartam | Flagyl | Metronidazol | Nirmet; (CN) China: Benzoylmetronidazole | Jia er na | Jia le ning | Metronidazole and glucose | Ya zhou ning | Yi fu qing; (CO) Colombia: Mecitro | Metronidazol | Metrozin | Nirmet | Otrozol | Sermetrol | Tenaflor; (CR) Costa Rica: Metronidazol | Metronidazol i.v | Metronidazol normon | Metronidazol pisa; (CZ) Czech Republic: Efloran | Flagyl | Metronidazol | Metronidazol avmc | Metronidazole noridem; (DE) Germany: Arilin | Clont | Flagyl | Fossyol | Infectoclont | Metronidazol | Metronidazol demo | Metronidazol Fresenius | Metronidazol noridem | Metronidazol saar | Vagimid; (DK) Denmark: Flagyl | Metronidazol Alpharma; (DO) Dominican Republic: Etron | Format | Imilox | Metotrex | Metrex | Metro | Metroan | Metrocaps | Metrocol | Metronidazol | Metronidazol alfa | Midazole | Otrozol | Servizol; (EC) Ecuador: Acromona | Amotein | Bectamet | Etron | Flagyl | Mebadiol | Metronidazol | Metroz | Nidazol | Nirmet | Otrozol | Servizol | Solucion de metronidazol | Tenaflor | Vetisol; (EE) Estonia: Efloran | Flagyl | Klion | Metronidazol | Metronidazol braun | Metronidazol nd | Metronidazol ns | Supplin; (EG) Egypt: Amrizole | Bumexazole | Copagaramon | Dumozol | Elyzol | Flagyl | Flazol | Metroflag | Trichogyl; (ES) Spain: Amotein | Flagyl | Metronidazol aurovitas | Metronidazol braun | Metronidazol cepa | Metronidazol Diasa | Metronidazol G.E.S. | Metronidazol normon; (ET) Ethiopia: Metris | Metrogyl | Polgyl; (FI) Finland: Flagyl | Metronidazol Claris | Metronidazole Vioser | Trikozol; (FR) France: Flagyl | Metronidazole arrow | Metronidazole baxter | Metronidazole bieffe | Metronidazole biogaran | Metronidazole dakota | Metronidazole fresenius | Metronidazole Lavoisier | Metronidazole merck | Metronidazole noridem | Metronidazole PCH | Metronidazole substiphar; (GB) United Kingdom: Elyzol | Metrolyl | Metrovex | Metrozol | Zadstat; (GR) Greece: Colpocin t | Flagyl | Gnostol | Metrogyl | Metronidazole cooper | Metronidazole/baxter | Metronidazole/Norma; (GT) Guatemala: Barmicil iv | Bectamet | Nirmet | Tenaflor; (HK) Hong Kong: Flagyl | Frothin | Fuzuin | Metronil | Tolizole; (HR) Croatia: Efloran | Medazol | Metronidazol B. Braun; (HU) Hungary: Flagyl | Klion | Metronidazol noridem | Metronidazol normon | Metronidazol teva | Metronidazol-human | Supplin; (ID) Indonesia: Anmerob | Biatron | Dazolin | Diazole | Dimedazol | Dumozol | Elyzol | Farnat | Fladex | Flagyl | Fortagyl | Metrofusin | Metronidazole fresenius | Promuba | Rindozol | Robukil | Ronazol | Supplin | Trichodazol | Trogyl | Vadazol | Velazol; (IE) Ireland: Flagyl | Metrovex; (IL) Israel: Flagyl; (IN) India: Aldezole | Altrogyl | Anaerid.s.i.v. | Aristogyl | Balgyl | Biozole | Flagyl | Flagyl er | Glucogyl | Lupigyl | Met | Metgyl | Metris | Metro | Metrocip | Metrogyl | Metrogyl er | Metrokem | Metron | Metronidazole & dextrose | Premogyl | Rogyl | Unimezol | Wingyl | Wometro; (IQ) Iraq: Flagyrol fort | Furazol forte | Metrobac | Safagyl; (IS) Iceland: Flagyl; (IT) Italy: Deflamon | Metronidazolo | Metronidazolo Aurobindo | Metronidazolo Bbu | Metronidazolo bieffe medital | Metronidazolo bioindustria | Metronidazolo kabi | Metronidazolo salif | Vagilen; (JO) Jordan: Colpocin t | Elyzol | Entogyl | Flagyl

| Fungizol | Metrazine | Metrolag | Supplin; (JP) Japan: Anaemetro | Asuzol | Flagyl | Kyometol | Meronidal | Metronidazole amel | Metronidazole merck hoei | Metronidazole nichiiiko | Nida | Salandol | Sawagyl | Takimetol | Trichocide | Wagitran; (KE) Kenya: Abagyl | Axagyl | Biozole | Flagyl | Metlife | Metris | Metrocach | Metrogyl | Metrolag | Metrozol | Metzole | Nirmet | Trichazole | Trichogyl | Trogyl | Troz; (KR) Korea, Republic of: Baxter metronidazole | Flagyl | Inno.n metronidazol | Metazole | Metrinal | Trizele | Tronal; (KW) Kuwait: Elyzol | Flagyl | Metronidazol | Metronidazole Braun | Mirazol | Neogyl | Supplin; (LB) Lebanon: Alfa metronidazole | Flagyl | Metris | Metronidazole fresenius | Supplin; (LT) Lithuania: Arilin | Eflora | Efloran | Flagyl | Klion | Metrogyl | Metronidasoli | Metronidazol | Metronidazol arena | Metronidazol bp | Metronidazole Vioser | Metroxan | Supplin; (LU) Luxembourg: Flagyl | Metronidazol; (LV) Latvia: Efloran | Flagyl | Flazole | Klion | Metrogyl | Metronidazol | Metronidazole fresenius | Metronidazolium | Triacide; (MA) Morocco: Flagyl | Zyrdol; (MX) Mexico: Amiyodazol | Flagenase | Flagyl | Flaxtec | Hasolmedel | Kabizol | Metronidazol | Metronidazol gi pi | Metronidazol rimsa | Metronidazol Tecnofarma | Otrozol | Servizol | Solsensi | Tenaflor | Vertisal; (MY) Malaysia: Flagyl | Frothin | Metagyl | Metrogyl | Metronidazole Pharmaniaga | Ranigyl; (NG) Nigeria: Amrizole | Cedazole | Flagyl | Jugyl | Kinglion metronidazole | Metrolife | Nci metronidazole | Nelb metronidazole | Nirmet | Rebok metronidazole | St dennis metronidazole | Trimetro | Troz | Unigyl; (NL) Netherlands: Elyzol | Flagyl | Metronidazol | Metronidazol Alpharma | Metronidazol aurobindo | Metronidazol noridem | Metronidazol pch; (NO) Norway: Flagyl | Metronidazol; (NZ) New Zealand: Aft Metronidazole | Colpocin t | Flagyl; (PA) Panama: Acumet | Metris | Metronidazol | Metronidazol b.braun | Metronidazol normon | Otrozol | Tricozol; (PE) Peru: Flagimed | Flagyl | Metrexol | Metronidazol | Metrostat | Metrozol; (PH) Philippines: Ailyzole | Altozol | Anerobia | Anerobizol | Antizol | Astrozol | Axagyl | B braun metronidazole | Bacimex | Cetaz | Dazomet | Eastmezol | Elizol | Erzol | Flagyl | Frothin | Godazole | Hizon metronidazole | Imetron | Medizole | Metdazid | Metrazol | Metrinol | Metroclear | Metromax | Metromed | Metronidazol | Metronidazol Winthrop | Microzol | Moniliasol | Neorobic | Newmet | Nicozole | Nidazoprime | Nirmet | Norstene | Nozol | Onizole | Panazole | Patryl | Pharex metronidazole | Promeddazole | Protozole | Servizol | Triconex | Tridel | Trozodaz | Usa-metronidazole | Zol | Zolegyl | Zolemet | Zolvex; (PK) Pakistan: Abozole | Amezole | Amorob | Anarob | Diazol | Diazole | Efloran | Flagyl | Flagynase | Flazol | Flezole | Gee metro | Gramex | Inglee | Intobix | Klint | Livgyl | Lomet | Markgyl | Medigyl | Metason | Metical | Metrinda | Metro | Metroate | Metrodan | Metrofit | Metrogyl | Metrolite | Metrome | Metrosol | Metrostat | Metrozil | Metrozine | Metwise | Metzeena | Metzin | Metzol | Mezole | Microgin | Mtz | Murazol | Narobe | Nidagyl | Opozol | Otsuzol | Palmizol | Protoxil | Prozyol | Resgyl | Revgyl | Saydazole | Sidazole | Trichazole | Troz | Zavad | Zonid; (PL) Poland: Efloran | Flagyl | Metronidazol | Metronidazol aurovitae | Metronidazol polpharma | Metronidazole b braun; (PR) Puerto Rico: Flagyl | Flagyl er | Metronidazole; (PT) Portugal: Dumozol | Metronidazol | Metronidazol basi | Metronidazol Generis | Metronidazol labesfal; (PY) Paraguay: Cariodent zol | Medazol | Metrobiot | Metronac | Metronidazol dutric | Metronidazol halexistar | Metronidazol phi | Metronidazol prosalud | Metrosol | Tricolan | Tricoval; (QA) Qatar: Apo-Metronidazole | Flagyl | Flagyl S | Flanzol | Fungizol | Metgyl | Metris | Metrolag | Metzol | Metzol-S | Negazole | Nida-Q | Supplin | Suprazole; (RO) Romania: Flagyl | Klion | Metronidazol arena | Metronidazol Vioser; (RU) Russian Federation: Alimezole | Aquametro | Bacimex | Edgil | Efloran | Flagyl | Klion | Metrogyl | Metrolacare | Metron | Metronidal | Metronidazole advance | Metronidazole escom | Metronidazole nycomed | Nirmet | Trichazol | Trichopol; (SA) Saudi Arabia: Elyzol | Flagyl | Mopizale | Nidazole | Riazole; (SE) Sweden: Flagyl | Metronidazol actavis | Metronidazol baxter | Metronidazol braun | Metronidazol eql pharma | Metronidazol norcox | Metronidazole Braun; (SG) Singapore: Flagyl | Ranigyl; (SI) Slovenia: Efloran | Flagyl | Metronidazol | Metronidazol braun; (SK) Slovakia: Efloran | Klion | Metronidazol braun | Metronidazole noridem; (SL) Sierra Leone: Bactazole; (SR) Suriname: Metronidazol | Metronidazol aurobindo; (SV) El Salvador: Otrozol; (TH) Thailand: Asiazole | Biogyl | Chanenidazole | Flagyl |

Klion | Klont | Lidazole | Manodazole | Medazyl | Medizole | Mepagyl | Metrazole | Metrex | Metrogyl | Metrolex | Metron | Metronid | Metronidazole fresenius kabi | Metronidazole iv b braun | Metrovid | Metrowin | Mezole | Milanidazole | Mydazole | Nida zole | Nirmet | Novamet | Octec | Temonas | Trichobon | Tridazol | Vagil; (TN) Tunisia: Anaeryl | Flagyl | Metrogyl | Metronidazol Biosedra | Metronidazole infomed | Nidazol | Protogyl | Protozol | Tamizol | Trichex; (TR) Turkey: Fladazol | Flagyl | Laronium | Metronidazol | Metrosele | Metrosol | Neofleks metronidazol | Nidazol | Polgyl; (TW) Taiwan: Anegyn | Antimonas | Arcazol | Chomozin | Chozine | Destrinalis | Flagyl | Frodin | Fulikan | Fulon | Fuzuin | Geel | Lifemiss | Medazole | Metoyui | Metricho | Metrocide | Metrodin | Metrole | Metron | Metrozole | Paurapol | Pynal | Sabs | Santon | Servizol | Terico-s | Tolizole | Tonilin | Tricogyl | Tricon; (UA) Ukraine: Efloran | Flagyl | Klion | Metressa | Metrid | Metrogyl | Metroxan | Metrozol | Trikasaid; (UG) Uganda: Aartrogyl | Abgyl | Axagyl | Glomet | Metlife | Metris | Metrogyl | Metron | Metroren fc | Metrozole | Nirmet | Zifamet; (UY) Uruguay: Dazolston | Flagyl | Ginplus | Metronidazol | Metronidazol B. Braun | Metronidazol Lazar | Metronidazol rivero | Metroplex | Noxidazol | Servizol | Setixol | Trimstat; (VE) Venezuela, Bolivarian Republic of: Bactrizol | Flegyl | Metris | Metrocof | Metrogyl | Metronidazol | Metrovax | Metrowell | Metrozol; (VN) Viet Nam: Aldezil | Moretel; (ZA) South Africa: Bio metronidazole | Flagyl | Metagyl | Metris | Metronidazole b braun | Metronidazole fresenius | Narobic | Sabax metronidazole | Trichazole; (ZM) Zambia: Flazole | Hecmetro 200 | Intramed trichazole | Metlife | Metris | Shalmet | Trichogyl; (ZW) Zimbabwe: Metris | Nirmet | Trichazole

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